Daniel Aarao Reis Arturi

Research Engineer and Software Engineer

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PROFESSIONAL SUMMARY

Research engineer and software engineer building agentic systems, cross-system benchmarks, and production-minded research infrastructure. Experience spanning LLM agents, filtered vector search, stopping-policy synthesis, full-stack systems, and reproducible performance evaluation.

EXPERIENCE

NYU Tandon School of Engineering, VIDA Center — May 2026-Present

Research Scientist | Agentic Systems and Research Infrastructure | Lea

- Developed Lea's project-aware agent harness with configurable sub-agents, Lean checking, and persistent proof history; first-author paper accepted to NeurIPS 2026 MATH-AI.
- Built the React/Vite/TypeScript client and Python/FastAPI/SQLite backend path, including typed server-sent events, live proof updates, shared state, usage tracking, and spend controls.
- Developed the Overleaf-to-Lean workflow across a Chrome extension and local Node.js companion, including dependency parsing, inline progress, manual review, counterexample handling, self-repair, and GitHub export.

New York University, Courant Institute of Mathematical Sciences — Sep 2026-Present

Independent Study Researcher | Filtered Vector Search | Advisor: Prof. Jinyang Li

- Developed FANNAgent, an offline LLM-guided system for workload-specific early stopping in filtered approximate nearest-neighbor search; implemented bounded policies inside pgvector's HNSW search without query-time model calls.
- Built FANNBench across PostgreSQL/pgvector, Elasticsearch/Lucene, and Weaviate, with exact filtered top-*k* references and native instrumentation separating search, feature-collection, and policy-execution costs.
- Measured 59.7-62.0% lower mean latency than native pgvector for an agent-generated policy on 300 development query-filter cases, with 95% mean Recall@10 versus 99% for the baseline.

McGill University and University of Passau — Aug 2025-Mar 2026

Research Intern | Graph Database Systems | Artifact | Earlier preprint

- Implemented Dewey and PrePost structural indexes for tree-shaped property-graph data and built a Dockerized benchmark across Apache AGE, Kuzu, and Neo4j using 100,000-node synthetic trees and LDBC SNB workloads.
- Measured 6.7x to more than 300,000x microbenchmark speedups and 1.1x to 16x gains on three LDBC queries in AGE, documenting mixed native-engine results and index-maintenance costs; co-authored the accepted EDBT 2027 vision paper.

McGill University, Distributed Information Systems Group — Jan-Jun 2026

Research Intern | Quantum Computing for Data Systems | Paper | Artifact

- Built a reproducible benchmark for distributed data allocation across exact ILP and brute-force oracles, D-Wave simulated annealing, hybrid CQM, Advantage hardware, and QAOA.
- Implemented resumable experiment runners and analysis-ready metrics for validity, constraint violations, optimality gap, embedding size, chain breaks, and timing while separating QPU access time from end-to-end cloud latency; the resulting first-author paper was accepted at VLDB 2026 QC&DKM and delivered as an oral presentation.

Algoverse AI Research — Apr 2025-Present

Research Intern | AI Safety and Mechanistic Interpretability | Workshop paper | Code

- Analyzed LoRA-adapted Qwen2.5-7B and Llama 3.1-8B models using PyTorch, parameter-subspace analysis, and linear mode connectivity.
- Built fine-tuning and evaluation pipelines for matched benign controls, data provenance, and post-hoc interventions; contributed to a full-length manuscript under review at ICLR 2027.

SELECTED ENGINEERING PROJECTS

NYC Housing Research

Python, FastAPI, SQLite, local-first software, macOS | [Code]

- Built a local-first legal research application over official NYC housing sources with full-text and semantic retrieval, evidence-cited answers, provider-cost controls, resilient backup and upgrade recovery, and a self-contained macOS desktop build.

Paxos Treasure Island

Java, Paxos, UDP, distributed systems | [Code]

- Implemented single-decree Paxos over UDP for total-order broadcast, with concurrent proposer handling, six crash-injection points, scripted consensus verification, and an interactive protocol-trace visualizer.

SELECTED RESEARCH OUTPUT

NeurIPS 2026 MATH-AI: accepted Lea paper, first author | **EDBT 2027**: accepted paper, co-author | **ICLR 2027**: full-length manuscript under review | **NeurIPS 2025 workshops**: joint-first-author paper, UniReps oral + honorable mention, MechInterp spotlight | **VLDB 2026 QC&DKM**: accepted paper, first author and oral presenter

EDUCATION

New York University, Courant Institute of Mathematical Sciences | M.S. Computer Science | Fall 2026-Present
McGill University | B.Sc. Honours Software Engineering | May 2026 | GPA 3.99/4.0 | First Class Honours

TECHNICAL SKILLS

Languages Python, Java, C/C++, SQL, TypeScript, JavaScript, Bash, Swift, Lean
Agents and ML PyTorch, Transformers, LoRA/PEFT, agent orchestration, LLM tooling, evaluation
Data and search PostgreSQL, pgvector, Elasticsearch/Lucene, Weaviate, HNSW, SQLite, Apache AGE, Kuzu, Neo4j
Infrastructure FastAPI, React, Docker, Git, GitHub Actions, Linux, Slurm, Jupyter